

“FACTS has been recognised as one of the key technologies of the decade that will shape the future”

India is doing very well with hydro power generation and moving in the right direction for other sources, including wind and solar. However, to meet the increasing consumer demands, the road ahead is still too long, feels Dr Fahd Hashiesh, fellow, Institution of Engineering and Technology and technology manager, ABB Ltd, UK, as he talks with Abhishek Mutha of EFY. He also throws light on upcoming technologies and the challenges that need to be tackled in the renewable space



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Q. Could you shed light on upcoming power generation and storage technologies, considering the fact that we could face power shortage in the future. Will these be environment friendly?

Energy generation and storage will form an integral part in the near future with a lot of focus on renewable energy. Examples of this are wind and solar generation, both of which are not fully predictable and dispatchable. These have to be accompanied by storage to support the generation dips when wind is not blowing or during night time. Also, storage will play a very important role in the future in supporting the grid system inertia.

There are many sources of energy storage and many of them are environment friendly, such as hydro storage and flywheel storage. Some of the storage technologies are matured and have been in use for a long time, while others are still under development as they still have a smaller storage capacity.

Q. What challenges does one face while integrating different sources of energy generation and storage to, say, a mega grid or the national power grid of a country?

Renewable energy will open the door to a lot of challenges that include power quality because of the electronic inverters and converters used with these technologies, loss of inertia, rate of change of frequency, fault ride through and reactive power balance. All of these will add new challenges, in addition to the optimal allocation for the storage unit for the system operator.

Q. Could you share an overview of the renewable sector in India today?

There is no doubt about the growth of energy generation in India over the past few years, especially renewable energy. This growth has placed India at the sixth position

worldwide. India is doing very well with hydro power generation and moving in the right direction for other sources, including wind and solar. But honestly, the road is still too long, especially to meet the increasing consumers' demands. The challenges will be from both system generation and operation point of view. These challenges include, but are not limited to, huge investments, planning, projects management, engineering and training.

Q. Power grid transmits about 50 per cent of the total power generated in India on its transmission network. Is there a transmission technique that could result in energy savings for the country, and do you see it being implemented in India any time soon?

FACTS (flexible alternating current transmission system) has been recognised as one of the key technologies of the decade that will shape the future. FACTS is categorised in two ways—series components and shunt components. Series components, such as TCSC (thyristor controlled series capacitor), will help directly to increase the transmission capabilities of existing lines without the need of building new lines, whereas shunt components are needed in certain locations of the networks to solve the problem of reactive power balance and enhance the fault ride through characteristics. This is just one example of FACTS' capability, which has a direct effect on investment saving. India already uses TCSC for a couple of different applications, and there are many opportunities to deploy FACTS technologies.

Q. What are the latest trends in the power system sector?

Digital substations and a lot of WAMPAC (wide area monitoring protection and control) implementations are the latest trends—superconductors could also be one. ●